

Green's Theorem in the Plane

If $M(x, y)$ and $N(x, y)$ be continuous functions of x and y having continuous partial derivatives $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ in a region R of the xy -plane bounded by a closed curve C , then

$$\oint_C (M dx + N dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

where C is traversed in the clockwise direction.

* This theorem is useful for changing a line integral around a closed curve C into a double integral over the region R enclosed by C .

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\vec{\nabla} \times \vec{F}) \cdot \hat{k} dx dy$$

Q. verify Green's theorem in the plane for $\oint_C (3x^2 - 8y^2) dx + (4y - 6xy) dy$ where C is the boundary of the region defined by

(a) $y = \sqrt{x}$, $y = x^2$
or $y^2 = x$

(b) $x = 0$, $y = 0$, $x + y = 1$

Solⁿ.

→ (a) $y = \sqrt{x}$ i.e. $y^2 = x$ and $y = x^2$ are two parabolas intersecting at $O(0, 0)$ and $A(1, 1)$

there $M = 3x^2 - 8y^2$, $N = 4y - 6xy$

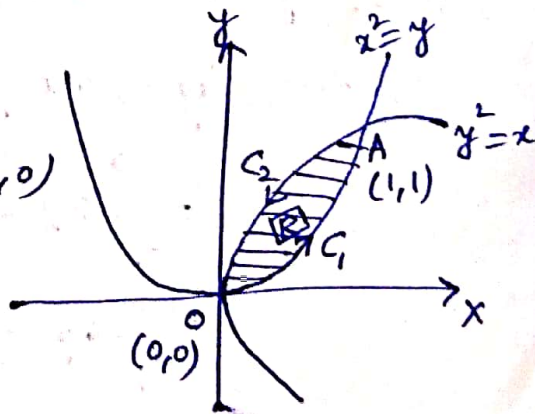
$$\frac{\partial M}{\partial y} = -16y, \quad \frac{\partial N}{\partial x} = -6y$$

$$\therefore \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = -6y - (-16y) = 10y$$

If R is the region bounded by C , then

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$= \int_0^1 \int_{x^2}^{\sqrt{x}} 10y dy dx = \int_0^1 5 [y^2]_{x^2}^{\sqrt{x}} dx$$



$$= 5 \int_0^1 (x - x^4) dx = 5 \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1$$

$$= 5 \left(\frac{1}{2} - \frac{1}{5} \right) = 5 \left(\frac{3}{10} \right) = \frac{3}{2} \quad \text{--- (1)}$$

$$\text{Also } \oint_C (M dx + N dy) = \int_C (M dx + N dy) + \int_{C_2} (M dx + N dy)$$

Along C_1 , $x^2 = y$, $\therefore 2x dx = dy$ and the limits of x are from 0 to 1

$$\therefore \text{Line integral along } C_1 = \int_{C_1} (M dx + N dy)$$

$$= \int_0^1 (3x^2 - 8x^4) dx + (4x^2 - 6x \cdot x^2) 2x dx$$

$$= \int_0^1 (3x^2 + 8x^3 - 20x^4) dx$$

$$= \left[x^3 + 2x^4 - 4x^5 \right]_0^1 = -1$$

Along C_2 , $y^2 = x$, $\therefore 2y dy = dx$ and the limits of y are from 1 to 0.

$$\therefore \text{Line integral along } C_2 = \int_{C_2} (M dx + N dy)$$

$$= \int_1^0 (3y^4 - 8y^2) 2y dy + (4y - 6y^2 \cdot y) dy$$

$$= \int_1^0 (4y - 22y^3 + 6y^5) dy = \left[2y^2 - \frac{11}{2}y^4 + y^6 \right]_1^0 = \frac{5}{2}$$

$$\therefore \text{Line integral along } C = -1 + \frac{5}{2} = \frac{3}{2}$$

$$\text{i.e., } \oint_C (M dx + N dy) = \frac{3}{2} \quad \text{--- (2)}$$

The equality of (1) and (2) verifies Green's theorem in the plane.

$$M = 3x^2 - 8y^2, N = 4y - 6xy$$

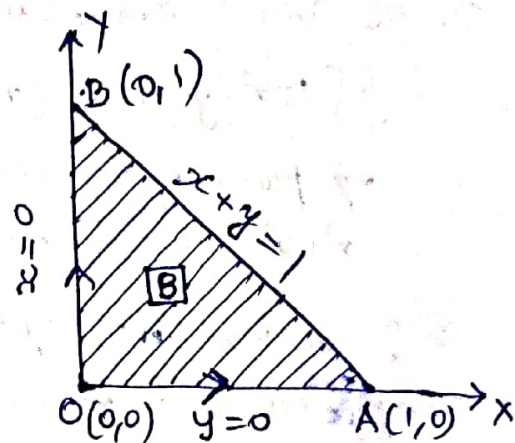
$$(b) x=0, y=0, x+y=1 \text{ or } y=1-x$$

$$\text{Here } \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$= \int_0^1 \int_0^{1-x} [-6y - (-16y)] dx dy$$

$$= \int_0^1 \int_0^{1-x} 10y dy dx = \int_0^1 5[y^2]_0^{1-x} dx$$

$$= 5 \int_0^1 (1-x)^2 dx = 5 \left[\frac{(1-x)^3}{-3} \right]_0^1 = -\frac{5}{3} (0-1) = \frac{5}{3} \quad \text{--- (1)}$$



Along OA, $y=0 \therefore dy=0$ and the limits of x are from 0 to 1

$$\therefore \text{Line integral along OA} = \int_0^1 3x^2 dx = [x^3]_0^1 \quad \left| \begin{array}{l} \int (M dx + N dy) \\ \text{Along } y=0 \\ \int 3x^2 dx \end{array} \right.$$

Along AB, $y=1-x, \therefore dy = -dx$ and the limits of x are from 1 to 0

$$\therefore \text{Line integral along AB} = \int_0^1 [3x^2 - 8(1-x)^2] dx + [4(1-x) - 6x(1-x)] (-dx)$$

$$= \int_0^1 (3x^2 - 8 + 16x - 8x^2 - 4 + 4x + 6x - 6x^2) dx$$

$$= \int_0^1 (-12 + 26x - 11x^2) dx$$

$$= [-12x + 13x^2 - \frac{11}{3}x^3]_0^1 = -[-12 + 13 - \frac{11}{3}] = \frac{8}{3}$$

Along BO, $x=0 \therefore dx=0$ and the limits of y are from 1 to 0.

$$\therefore \text{Line integral along BO} = \int_0^1 4y dy = [2y^2]_1^0 = -2$$

$$\therefore \text{Line integral along C (i.e., along OABO)} = 1 + \frac{8}{3} - 2 = \frac{5}{3} \rightarrow$$

$$\text{i.e., } \oint_C (M dx + N dy) = \frac{5}{3} \quad \text{--- (2)}$$

The equality of (1) and (2) verifies Green's theorem in the plane.

Q. Use Green's theorem in a plane to evaluate the integral $\oint_C [(2x^2 - y^2)dx + (x^2 + y^2)dy]$ where C is the boundary in the xy -plane of the area enclosed by the axes and the semi-circle $x^2 + y^2 = 1$ in the upper half xy -plane.

Solⁿ. If R is the region bounded by the closed curve C , then by Green's theorem in the plane, we have

$$\oint_C (M dx + N dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Here $M = 2x^2 - y^2$, $N = x^2 + y^2$

$$\frac{\partial M}{\partial y} = -2y, \quad \frac{\partial N}{\partial y} = 2x$$

$$\therefore \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 2(x + y)$$

The region is bounded by

$$x = -1, x = 1, y = 0, y = \sqrt{1-x^2}$$

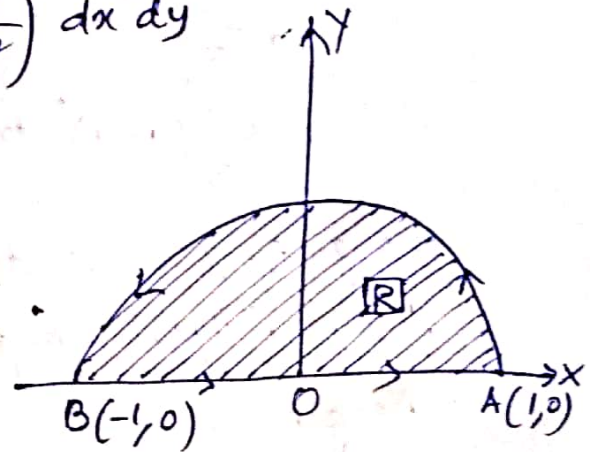
$$\therefore \oint_C [(2x^2 - y^2)dx + (x^2 + y^2)dy] = \iint_R 2(x + y) dx dy$$

$$= \int_{-1}^1 \int_0^{\sqrt{1-x^2}} 2(x + y) dy dx = 2 \int_{-1}^1 \left[xy + \frac{y^2}{2} \right]_0^{\sqrt{1-x^2}} dx$$

$$= 2 \int_{-1}^1 \left[x\sqrt{1-x^2} + \frac{1}{2}(1-x^2) \right] dx$$

$$= \int_{-1}^1 (1-x^2) dx, \text{ since } \int_{-1}^1 x\sqrt{1-x^2} dx = 0$$

$$= 2 \int_0^1 (1-x^2) dx = 2 \left[x - \frac{x^3}{3} \right]_0^1 = 2 \left(1 - \frac{1}{3} \right) = \frac{4}{3}$$



Q. Use Green's theorem to find the value of the line integral

$$\int_C \vec{F} \cdot d\vec{r} \text{ where } \vec{F} = y^3 \hat{i} - x^3 \hat{j} \text{ and } C: x^2 + y^2 = a^2, z=0$$

Solⁿ $\rightarrow$ Here $\vec{F} = y^3 \hat{i} - x^3 \hat{j} = M\hat{i} + N\hat{j}$ $\therefore M = y^3, N = -x^3$

By Green's theorem $\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$

where R is the plane area bounded by the closed curve $C: x^2 + y^2 = a^2, z=0$

$$\therefore \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \iint_R -3(x^2 + y^2) dx dy \text{ changing to polar coordinates}$$

$$= -3 \int_0^{2\pi} \int_0^a r^2 r dr d\theta = -3 \int_0^{2\pi} d\theta \int_0^a r^3 dr$$

$$= -\frac{3}{2} \pi a^4$$

Q. Evaluate by Green's theorem, $\int_C e^{-x} (\sin y dx + \cos y dy)$ where C is the rectangle with vertices $(0,0), (\pi,0), (\pi, \pi/2), (0, \pi/2)$

Solⁿ $\rightarrow$ By Green's theorem $\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$

Here $M = e^{-x} \sin y, N = e^{-x} \cos y$, where C is a closed curve enclosing the region R. In the given case

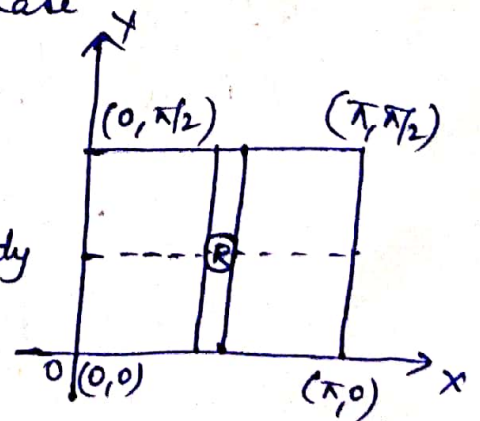
$$\therefore \int_C e^{-x} \sin y dx + e^{-x} \cos y dy$$

$$= \iint_R \left(\frac{\partial}{\partial x} (e^{-x} \cos y) - \frac{\partial}{\partial y} (e^{-x} \sin y) \right) dx dy$$

where R is the region enclosed in the rectangle with vertices $(0,0), (\pi,0), (\pi, \pi/2), (0, \pi/2)$

$$= \iint_R (-e^{-x} \cos y - e^{-x} \cos y) dy dx$$

$$= -2 \int_0^{\pi} e^{-x} dx \int_0^{\pi/2} \cos y dy = 2(e^{-\pi} - 1)$$



Stokes's Theorem.

Relⁿ between line and Surface Integral

If S be an open surface bounded by a closed curve C and $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ be any vector point fn having continuous first order partial derivatives, then $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$ where $\hat{n}$ is a unit normal vector at any point of S drawn in the sense in which a right handed screw would advance when rotated in the sense of description C .

Stokes Theorem can be written as

$$\oint_C (\vec{F}_1 dx + \vec{F}_2 dy + \vec{F}_3 dz)$$

$$= \iint_S \left[\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial x} \right) \cos \alpha + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \cos \beta + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \cos \gamma \right] ds$$

Let $\hat{n}$ make angles α, β, γ with +ve directions of x, y, z axis then

$$\hat{n} \cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}$$

Also $r = x\hat{i} + y\hat{j} + z\hat{k}$ so that $d\vec{r} = \hat{i}dx + \hat{j}dy + \hat{k}dz$

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

$$= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \hat{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \hat{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \hat{k}$$

$$\therefore \text{curl } \vec{F} \cdot \hat{n} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \cos \alpha + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \cos \beta + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \cos \gamma$$

$$\text{Also } \vec{F} \cdot d\vec{r} = (F_1\hat{i} + F_2\hat{j} + F_3\hat{k}) \cdot (\hat{i}dx + \hat{j}dy + \hat{k}dz) \\ = F_1 dx + F_2 dy + F_3 dz$$

Statement of Stokes's theorem.

* The line integral of the tangential component of a vector point fn $\vec{F}$ taken around a closed curve C is equal to the surface integral of the normal component of $\vec{F}$ taken over any surface S having C as its boundary.

* If S is an open surface (two sided) bounded by a closed non-intersecting curve C and if a vector field $\vec{F}$ has continuous partial derivatives, then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, ds = \iint_S (\nabla \times \vec{F}) \cdot d\vec{s}, \text{ where } C \text{ is traversed in positive direction}$$

Corollary: If two surfaces S_1 and S_2 have same bounding curve C ,

then $\iint_{S_1} (\nabla \times \vec{F}) \cdot d\vec{s} = \iint_{S_2} (\nabla \times \vec{F}) \cdot d\vec{s}$ since each is equal to $\oint_C \vec{F} \cdot d\vec{r}$

Q. Verify Stokes' Theorem for the function $\vec{F} = x^2\vec{i} + xy\vec{j}$ integrated round the square in the plane $z=0$ and bounded by the lines $x=0, y=0, x=a$ and $y=a$

Solⁿ → According to Stokes' theorem $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} ds$

Here $\vec{F} = x^2\vec{i} + xy\vec{j}$

$$\therefore \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & xy & 0 \end{vmatrix} = k(y-0) = ky$$

$$\text{Thus } \iint_S \nabla \times \vec{F} \cdot \hat{n} ds = \iint_S ky \cdot k dx dy = \int_0^a \int_0^a y dy dx = \frac{1}{2} a^3$$

$$\text{and } \oint_C \vec{F} \cdot d\vec{r} = \int_C (x^2\vec{i} + xy\vec{j}) \cdot (\vec{i} dx + \vec{j} dy) = \int_C x^2 dx + xy dy$$

where C is the closed curve $OABC$

Along $OA, y=0, x$ varies from 0 to a

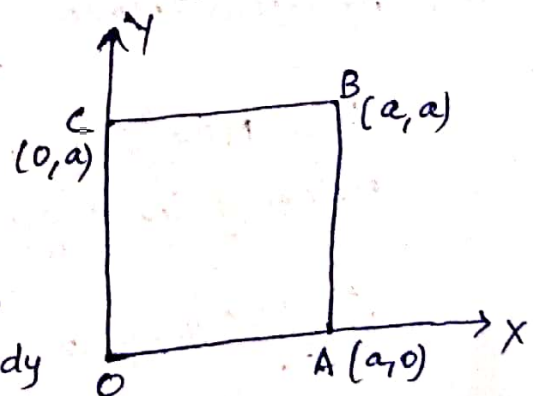
along $AB, x=a, y$ varies from 0 to a

along $BC, y=a$ and x varies from a to 0

$$\therefore \oint_C \vec{F} \cdot d\vec{r} = \int_0^a x^2 dx + \int_0^a ay dy + \int_a^0 x^2 dx + \int_a^0 0 dy$$

$$= \frac{a^3}{2}$$

Hence Stokes' theorem is verified.



Q. use Stokes's theorem to evaluate $\oint_C \vec{V} \cdot d\vec{r}$ where $\vec{V} = y^2\hat{i} + xy\hat{j} + xz\hat{k}$ and C is the bounding curve of the hemisphere $x^2 + y^2 + z^2 = 9, z \geq 0$ oriented in the positive direction

Solⁿ $\rightarrow$ C is the bounding curve of the hemisphere $x^2 + y^2 + z^2 = 9, z \geq 0$ and hence $C: x^2 + y^2 = 9, z = 0$
 $\vec{V} = y^2\hat{i} + xy\hat{j} + xz\hat{k}$. Using Stokes theorem we have

$\oint_C \vec{V} \cdot d\vec{r} = \iint_S \nabla \times \vec{V} \cdot d\vec{s}$ where S is the area under the curve C i.e., the area inside $x^2 + y^2 = 9$

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & xy & xz \end{vmatrix} = \hat{i}(0) + \hat{j}(0 - z) + \hat{k}(y - zy) = -z\hat{j} - y\hat{k}$$

$$\begin{aligned} \therefore \oint_C \vec{V} \cdot d\vec{r} &= \iint_S (-z\hat{j} - y\hat{k}) \cdot (-\hat{k}) dx dy = \iint_S y dy dx \\ &= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} y dy dx = 0 \end{aligned}$$

Q. Evaluate $\oint_C \vec{V} \cdot d\vec{r}$ using Stokes's theorem if

$\vec{V} = (3x + 2z)\hat{i} + (x + 3y)\hat{j} + (2y - 3z)\hat{k}$ and C is the curve of intersection of the plane $6x + 3y + 4z = 12$ with the coordinate planes.

Solⁿ $\rightarrow$ $\vec{V} = (3x + 2z)\hat{i} + (x + 3y)\hat{j} + (2y - 3z)\hat{k}$

$$\therefore \nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x+2z & x+3y & 2y-3z \end{vmatrix} = \hat{i}(2-0) + \hat{j}(2-0) + \hat{k}(1-0) = 2\hat{i} + 2\hat{j} + \hat{k}$$

If $\hat{n}$ is unit vector along the normal to the plane

$$6x+3y+4z=12, \text{ then } \hat{n} = \frac{6\hat{i}+3\hat{j}+4\hat{k}}{\sqrt{61}}$$

By Stokes's theorem, $\oint_C \vec{V} \cdot d\vec{r} = \iint_S (2\hat{i}+2\hat{j}+\hat{k}) \cdot \hat{n} dS$

$$\therefore \oint_C \vec{V} \cdot d\vec{r} = \iint_S \frac{(2\hat{i}+2\hat{j}+\hat{k}) \cdot (6\hat{i}+3\hat{j}+4\hat{k})}{\sqrt{61}} dS$$

$$= \frac{1}{\sqrt{61}} \iint_S (12+6+4) dS = \frac{22}{\sqrt{61}} \iint_S dS$$

The given plane is $6x+3y+4z=12$ or, $\frac{x}{2} + \frac{y}{4} + \frac{z}{3} = 1$

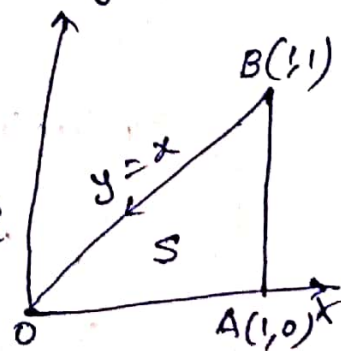
It meets the coordinate axis in $(2,0,0)$, $(0,4,0)$ and $(0,0,3)$

whose area is $\sqrt{61}$ $\therefore \oint_C \vec{V} \cdot d\vec{r} = 22$

Q. Evaluate $\oint_C \vec{V} \cdot d\vec{r}$ by Stokes's theorem, where $\vec{V} = y^2\hat{i} + x^2\hat{j} - (x+z)\hat{k}$ and C is the boundary of the triangle with vertices at $(0,0,0)$, $(1,0,0)$ and $(1,1,0)$

Solⁿ Since z -coordinates of each vertex of the triangle is zero, $\therefore$ the triangle lies in the xy -plane and $\hat{n} = \hat{k}$

$$\text{Curl } \vec{V} \text{ or } \nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & x^2 & -(x+z) \end{vmatrix} = \hat{j} + 2(x-y)\hat{k}$$



$$\therefore \nabla \times \vec{V} \cdot \hat{n} = [\hat{j} + 2(x-y)\hat{k}] \cdot \hat{k} = 2(x-y)$$

The eqn of line OB is $y=x$

By Stokes's theorem $\oint_C \vec{V} \cdot d\vec{r} = \iint_S \nabla \times \vec{V} \cdot \hat{n} dS$

$$= \int_0^1 \int_0^x 2(x-y) dy dx$$

$$= \int_0^1 2 \left[xy - \frac{y^2}{2} \right]_0^x dx = 2 \int_0^1 \left(x^2 - \frac{x^2}{2} \right) dx = \int_0^1 x^2 dx = \frac{1}{3}$$

Divergence Theorem or Gauss theorem
(Relⁿ betⁿ surface and volume integral).

If $\vec{F}$ is a vector point fn having continuous first order partial derivatives in the region V bounded by a closed surface S , then $\iiint_V \nabla \cdot \vec{F} dV = \iint_S \vec{F} \cdot \hat{n} ds$, where $\hat{n}$ is the outwards drawn unit normal vector to the surface S .

$$\left[(\vec{F} = F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}), \quad \nabla \cdot \vec{F} = \text{div } \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right]$$

Q. verify divergence theorem for $\vec{F} = x^2 \hat{i} + z \hat{j} + yz \hat{k}$, taken over the cube bounded by $x=0, x=1, y=0, y=1$.

Solⁿ → Divergence theorem states that

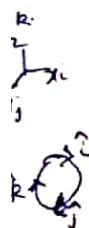
$$\int_V \text{div } \vec{F} \cdot dV = \int_S \vec{F} \cdot \hat{n} ds$$

$$\text{div } \vec{F} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(z) + \frac{\partial}{\partial z}(yz) = 2x + y$$

$$\int_V \text{div } \vec{F} \cdot dV = \int_0^1 \int_0^1 \int_0^1 (2x+y) dx dy dz = \int_0^1 \int_0^1 \left[\int_0^1 (2x+y) dx \right] dy dz$$

$$= \int_0^1 \int_0^1 [x^2 + xy]_0^1 dy dz = \int_0^1 \int_0^1 (1+y) dy dz$$

$$= \int_0^1 \left(y + \frac{y^2}{2} \right)_0^1 dz = \int_0^1 \frac{3}{2} dz = \frac{3}{2} \quad \text{--- (1)}$$



Now, let

$$S_1 = \text{face OABC}, z=0, \hat{n} = -\hat{k}$$

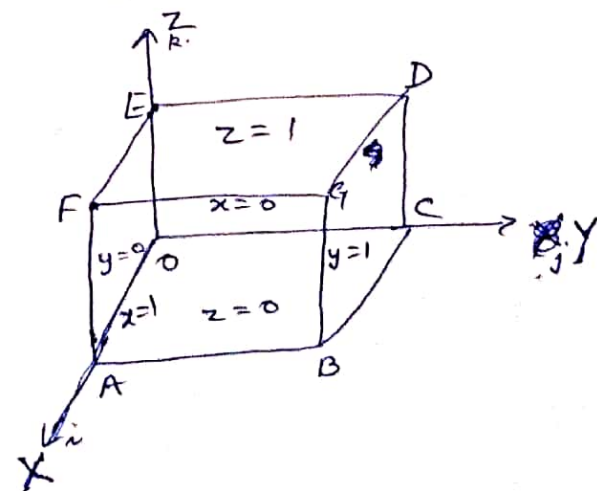
$$S_2 = \text{face DEFG}, z=1, \hat{n} = \hat{k}$$

$$S_3 = \text{face BCDG}, y=1, \hat{n} = \hat{j}$$

$$S_4 = \text{face OEFA}, y=0, \hat{n} = -\hat{j}$$

$$S_5 = \text{face OCDE}, x=0, \hat{n} = -\hat{i}$$

$$S_6 = \text{face ABGF}, x=1, \hat{n} = \hat{i}$$



$$\int_S \vec{F} \cdot \hat{n} ds = \int_{S_1} \vec{F} \cdot \hat{n} ds + \int_{S_2} \vec{F} \cdot \hat{n} ds + \int_{S_3} \vec{F} \cdot \hat{n} ds + \int_{S_4} \vec{F} \cdot \hat{n} ds + \int_{S_5} \vec{F} \cdot \hat{n} ds + \int_{S_6} \vec{F} \cdot \hat{n} ds$$

$$\text{Now, } \int_{S_1} \vec{F} \cdot \hat{n} ds = \int_{S_1} (x^2 \hat{i} + z \hat{j} + yz \hat{k}) \cdot (-\hat{k}) dx dy = 0$$

$$\int_{S_2} \vec{F} \cdot \hat{n} dS = \int_0^1 \int_0^1 y dx dy = \frac{1}{2}, \quad \int_{S_3} \vec{F} \cdot \hat{n} dS = \int_0^1 \int_0^1 z dx dz = \frac{1}{2}$$

$$\int_{S_4} \vec{F} \cdot \hat{n} dS = \int_0^1 \int_0^1 -z dx dz = -\frac{1}{2}, \quad \int_{S_5} \vec{F} \cdot \hat{n} dS = 0$$

$$\int_{S_6} \vec{F} \cdot \hat{n} dS = \int_0^1 \int_0^1 dy dz = 1$$

$$\int_S \vec{F} \cdot \hat{n} dS = 0 + \frac{1}{2} + \frac{1}{2} - \frac{1}{2} + 0 + 1 = \frac{3}{2} \quad \text{--- (2)}$$

From (1) & (2) Divergence Theorem is verified.

Q. Verify the divergence theorem for the vector field $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ taken over the region bounded by the cylinder $x^2 + y^2 = 4$, $z = 0$, $z = 3$.

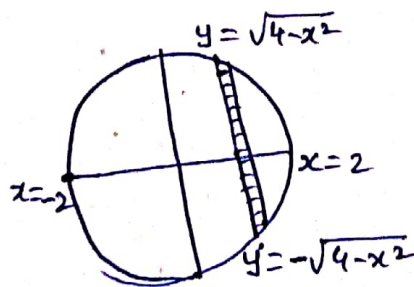
Solⁿ.

→ By Gauss Divergence theorem

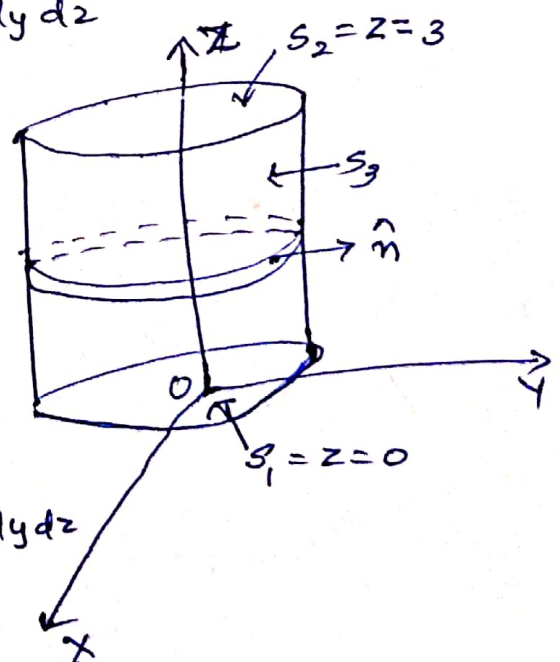
$$\int_S \vec{F} \cdot \hat{n} dS = \iiint_V \nabla \cdot \vec{F} dV. \quad \text{Here } \vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$$

$$\text{The volume integral} = \iiint_V \nabla \cdot \vec{F} dV = \iiint_V \left[\frac{\partial}{\partial x}(4x) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2) \right] dx dy dz$$

$$= \iiint_V (4 - 4y + 2z) dx dy dz$$



$$= \int_{x=-2}^2 \int_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^3 (4 - 4y + 2z) dx dy dz$$



$$= \int_{x=-2}^2 \int_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} [4z - 4yz + z^2]_0^3 dx dy$$

$$= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (21 - 12y) dy dx = 2 \int_{-2}^2 \int_0^{\sqrt{4-x^2}} 21 dy dx - 0 = 21 \times 4\pi$$

$$= 84\pi$$

$$\left[\because \text{area of circle } x^2 + y^2 = 4 \text{ in } xy\text{-plane} \right. \\ \left. = \pi(2)^2 = 4\pi \right]$$

$$\therefore \iiint_V \text{div } \vec{F} dv = 84\pi$$

The surface S consists of three surfaces, one being the base $S_1 (z=0)$ second being the top $S_2 (z=3)$ and third the curved surface S_3 of the cylinder $x^2 + y^2 = 4$ between $z=0$ and $z=3$.

Thus the surface integral $\cancel{S} = S_1$

$$= \iint_S \vec{F} \cdot \hat{n} ds = \iint_{S_1} \vec{F} \cdot \hat{n} ds_1 + \iint_{S_2} \vec{F} \cdot \hat{n} ds_2 + \iint_{S_3} \vec{F} \cdot \hat{n} ds_3$$

$$\text{On } S_1 (z=0), \hat{n} = -\hat{k} \text{ and } \vec{F} = 4x\hat{i} - 2y^2\hat{j} \therefore \vec{F} \cdot \hat{n} = 0$$

$$\therefore \iint_{S_1} \vec{F} \cdot \hat{n} ds_1 = 0$$

$$\text{On } S_2 (z=3), \hat{n} = \hat{k} \text{ and } \vec{F} = 4x\hat{i} - 2y^2\hat{j} + 9\hat{k} \therefore \vec{F} \cdot \hat{n} = 9$$

$$\therefore \iint_{S_2} \vec{F} \cdot \hat{n} ds_2 = 9 \iint_{S_2} ds_2 = 9(\pi(2)^2) = 36\pi \quad \left\{ \begin{array}{l} ds = dx dy \end{array} \right.$$

On S_3 , the curved surface of the cylinder ($x^2 + y^2 = 4$), the normal to $x^2 + y^2 = 4$ is in the direction of $\nabla(x^2 + y^2) = 2x\hat{i} + 2y\hat{j}$ ($\because \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(y^2) = 2x\hat{i} + 2y\hat{j}$)

$\therefore$ The unit vector $\hat{n}$ along the normal to the curved surface is given by

$$\hat{n} = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{4x^2 + 4y^2}} = \frac{2(x\hat{i} + y\hat{j})}{2\sqrt{x^2 + y^2}} = \frac{x\hat{i} + y\hat{j}}{2} \quad (\text{as } x^2 + y^2 = 4)$$

$$\text{hence } \vec{F} \cdot \hat{n} = (4x\hat{i} - 2y^2\hat{j} + 2\hat{k}) \cdot \left(\frac{x\hat{i} + y\hat{j}}{2} \right) = \frac{4x^2 - 2y^3 + 2x^2 - y^3}{2}$$

$$\therefore \iint_{S_3} \vec{F} \cdot \hat{n} dS_3 = \iint_{S_3} (2x^2 - y^3) dS_3$$

Since S_3 is a circle $x^2 + y^2 = 4$, so taking its polar form

$$x = 2 \cos \theta, y = 2 \sin \theta \quad (\text{here } r=2)$$

$$\text{or } x = 2 \cos \theta, y = 2 \sin \theta \quad \text{and } dS_3 = 2 d\theta dz$$

$$\text{and we get } \iint_{S_3} \vec{F} \cdot \hat{n} dS_3 = \int_{\theta=0}^{2\pi} \int_{z=0}^3 [2(2 \cos \theta)^2 - (2 \sin \theta)^3] 2 dz d\theta$$

$$= 48 \int_0^{2\pi} (\cos^2 \theta - \sin^3 \theta) d\theta = 48\pi + 0$$

Thus the given surface integral $= 0 + 36\pi + 48\pi = 84\pi$
which is equal to the volume integral,
which verifies divergence theorem.

Q. Find $\iint_S \vec{F} \cdot \hat{n} dS$ where $\vec{F} = (2x + 3z)\hat{i} - (xz + y)\hat{j} + (y^2 + 2z)\hat{k}$
and S is the surface of the sphere having centre at $(3, -1, 2)$
and radius 3.

Solⁿ. Here S is the surface of the sphere $(x-3)^2 + (y+1)^2 + (z-2)^2 = 9$
Unit vector $\hat{n}$ along the normal to the above surface, is

$$\text{given by } \hat{n} = \frac{2(x-3)\hat{i} + 2(y+1)\hat{j} + 2(z-2)\hat{k}}{\sqrt{4(x-3)^2 + 4(y+1)^2 + 4(z-2)^2}} = \frac{1}{3}[(x-3)\hat{i} + (y+1)\hat{j} + (z-2)\hat{k}]$$

$$\left[\hat{n} = \frac{\text{grad } \phi}{|\text{grad } \phi|}, \quad \text{let } \phi = (x-3)^2 + (y+1)^2 + (z-2)^2 - 9 = 0 \right.$$

$$\left. \text{grad } \phi = \hat{i} \frac{\partial}{\partial x} (x-3)^2 + \hat{j} \frac{\partial}{\partial y} (y+1)^2 + \hat{k} \frac{\partial}{\partial z} (z-2)^2 \right.$$

$$\left. = 2(x-3)\hat{i} + 2(y+1)\hat{j} + 2(z-2)\hat{k} \right]$$

$$\text{Here } \vec{F} = (2x + 3z)\hat{i} - (xz + y)\hat{j} + (y^2 + 2z)\hat{k}$$

$\therefore S$ is a closed surface, we can use divergence theorem

$$\text{and get } \iint_S \vec{F} \cdot \hat{n} dS = \iiint_V \nabla \cdot \vec{F} dV = \iiint_V \left[\frac{\partial}{\partial x} (2x + 3z) - \frac{\partial}{\partial y} (xz + y) + \frac{\partial}{\partial z} (y^2 + 2z) \right] dx dy dz$$

$$= \iiint_V (2 - 1 + 2) dx dy dz = 3 \times \text{volume of the sphere}$$

$$= 3 \cdot \frac{4}{3} \pi 3^3 = 108\pi$$

Q. Use Gauss divergence theorem to evaluate $\iint_S (\vec{F} \cdot \hat{n}) ds$ where $\vec{F} = x^2 z \hat{i} + y \hat{j} - x z^2 \hat{k}$ and S is the bounding of the region bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4y$.

Solⁿ $\rightarrow \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V (\nabla \cdot \vec{F}) dV = \iiint_V (2xz + 1 - 2xz) dV$

$$= \iiint_V dV = \int_{y=0}^4 \int_{x=-\sqrt{4y-y^2}}^{\sqrt{4y-y^2}} \int_{z=x^2+y^2}^{4y} dz dx dy$$

(because the projection of S on the xy plane is $x^2 + y^2 = 4y$)

$$= \int_{y=0}^4 \int_{x=-\sqrt{4y-y^2}}^{\sqrt{4y-y^2}} (4y - x^2 - y^2) dx dy$$

$$= 2 \int_{y=0}^4 \left[x(4y - y^2) - \frac{x^3}{3} \right]_0^{\sqrt{4y-y^2}} dy$$

$$= 2 \int_0^4 \left[(4y - y^2)^{3/2} - \frac{1}{3} (4y - y^2)^{3/2} \right] dy$$

$$= \frac{4}{3} \int_0^4 (4y - y^2)^{3/2} dy$$

$$= \frac{4}{3} \int_0^4 [4 - (y-2)^2]^{3/2} dy = \frac{64}{3} \int_0^{\pi/2} \cos^4 t dt = \frac{64}{3} \cdot \frac{3}{8} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = 12\pi$$

$\left(\begin{matrix} n = \text{even} = 4 \\ \int_0^{\pi/2} \cos^n t dt = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdot \dots \cdot \frac{\pi}{2} \end{matrix} \right)$
 $4 = \frac{64 \cdot 3 \cdot 1 \cdot \pi}{4 \cdot 2 \cdot 2} = 12\pi$